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ORIGINAL ARTICLE

Experimental and numerical studies of re-stirring and re-heating effects on mechanical properties in friction stir additive manufacturing

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Abstract
Friction stir additive manufacturing is studied experimentally and numerically. Different to conventional friction stir welding, re-stirring and re-heating can be found *for* friction stir additive manufacturing. The new generated phenomena are investigated by new integrated model, which is established based on Monte Carlo method and precipitate evolution model with combination of finite element model. Corresponding experiments are performed. Results indicate that different grain morphology and size, different hardness, and different yield strength can be found in different layers. The re-stirring and re-heating to the formed layers can lead to enhancement of mechanical properties with smaller grains and with higher hardness and yield strength. In comparison with first layer, the hardness and yield strength on second layer are increased due to the decrease of the peak temperature. After 2–3 layers, the mechanical properties of formed layers are not further affected by the new added layers.

Keywords Additive manufacturing · Friction stirring · Precipitates · Recrystallization · Re-stirring · Re-heating

1 Introduction

Friction stir additive manufacturing (FSAM) is a new additive manufacturing technology in solid state developed based on friction stir welding (FSW) [1, 2]. In comparison with laser additive manufacturing, no melting occurs in FSAM. So, thermal residual distortion can be minimized. Recrystallization instead of solidification becomes the main mechanism for production by FSAM.

Palanivel et al. [3] built a multilayered stack of Mg-based WE43 alloy by FSAM to investigate microstructural evolutions and structural performances. It was found that the recrystallization leads to finer grain sizes (2–3 μm). Mao et al. [4] used FSAM to produce Al-based components and found that the grain sizes can be different along the *build* direction. In comparison with Al substrate, both the tensile strength and the elongation are increased by FSAM. Yu et al. [5] recognized FSAM as non-beam metal additive manufacturing and believed FSAM can be used to solve the problems occurring in beam-based processes, e.g., solidification issues, anisotropic mechanical properties, and restrictions on powder attributes. Sharma et al. [6] fabricated tailor-made functionally graded composites (Al + TiC) by FSAM. The mechanical test results are correlated with the electron backscatter diffraction analysis for the investigation on process mechanism. The existing initial works on FSAM show its potential applications on industries and its advantages in comparison to beam-based additive manufacturing.

In comparison with FSW, re-stirring and re-heating can be found in FSAM. When the new layer is added in FSAM, the effect of the manufacturing of current layer on the former layers needs to be investigated for controlling of microstructures and mechanical properties for this attractive new additive manufacturing technology. Numerical method with combination of experiment is the most appropriate tool for research on mechanism of

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manufacturing processes. The main methods on simulation of microstructural evolutions include the Monte Carlo (MC) method [7], cellular automaton (CA) method [8], and phase field method [9], which can be also usually used in simulation of solidifications and phase transformations in beam- or non-beam-based additive manufacturing. Zhang et al. [10] compared the three mentioned numerical methods in simulation of microstructural evolutions in additive manufacturing. For use of Monte Carlo method and cellular automation method, physical meaning should be pre-defined. Zhang et al. [11] established a relation between real temperature history and Monte Carlo step to simulate the grain coarsening in the welding zones in FSW. Zhang and Hu [12] extended this model to 3D case. Wu and Zhang [13] further considered the effect of precipitates *on* grain growth in FSW. It was found that the grain sizes can be decreased by the added precipitates in MC model. As a precipitate strengthened material, numerical simulation of precipitate evolutions and the precipitate effects on grain growth are very important for further prediction on mechanical properties in FSW/FSAM of aluminum alloy. Zhang et al. [14] established the precipitate evolution model of FSW to investigate the precipitate evolutions in the FSW process and try to explain the effect of post weld heat treatment on improvement of mechanical properties. Cellular automaton method can be also combined with process model of FSW to simulate the grain evolutions in FSW. Akbari et al. [15] combined cellular automaton model and the Kocks-Mecking and Laasraoui-Jonas model to simulate the grain size distributions in FSW. Valvi et al. [16] developed cellular automata finite element approach for prediction on grain sizes of both similar and dissimilar FSWed metals. Shojaeefard et al. [17] combined cellular automaton and Taguchi method for optimization of microstructural and mechanical properties in FSW. Song et al. [18] further developed cellular automaton model to study the recrystallization in FSW of titanium alloy.

The precipitate evolution can affect the grain growth and is the key factor for controlling of mechanical properties. The combination of precipitate evolution model and Monte Carlo model of recrystallization provides the possibility for integrated modeling of microstructures and mechanical properties in FSAM as well as FSW. Re-stirring and re-heating makes FSAM being different to FSW. The integrated modeling of microstructures (recrystallization and precipitates) and mechanical properties, as well as the experiments of FSAM, can provide detailed insight into the mechanism for controlling of microstructures and the evaluations of the re-stirring and re-heating effects on FSAM, which is the motivation of current work.

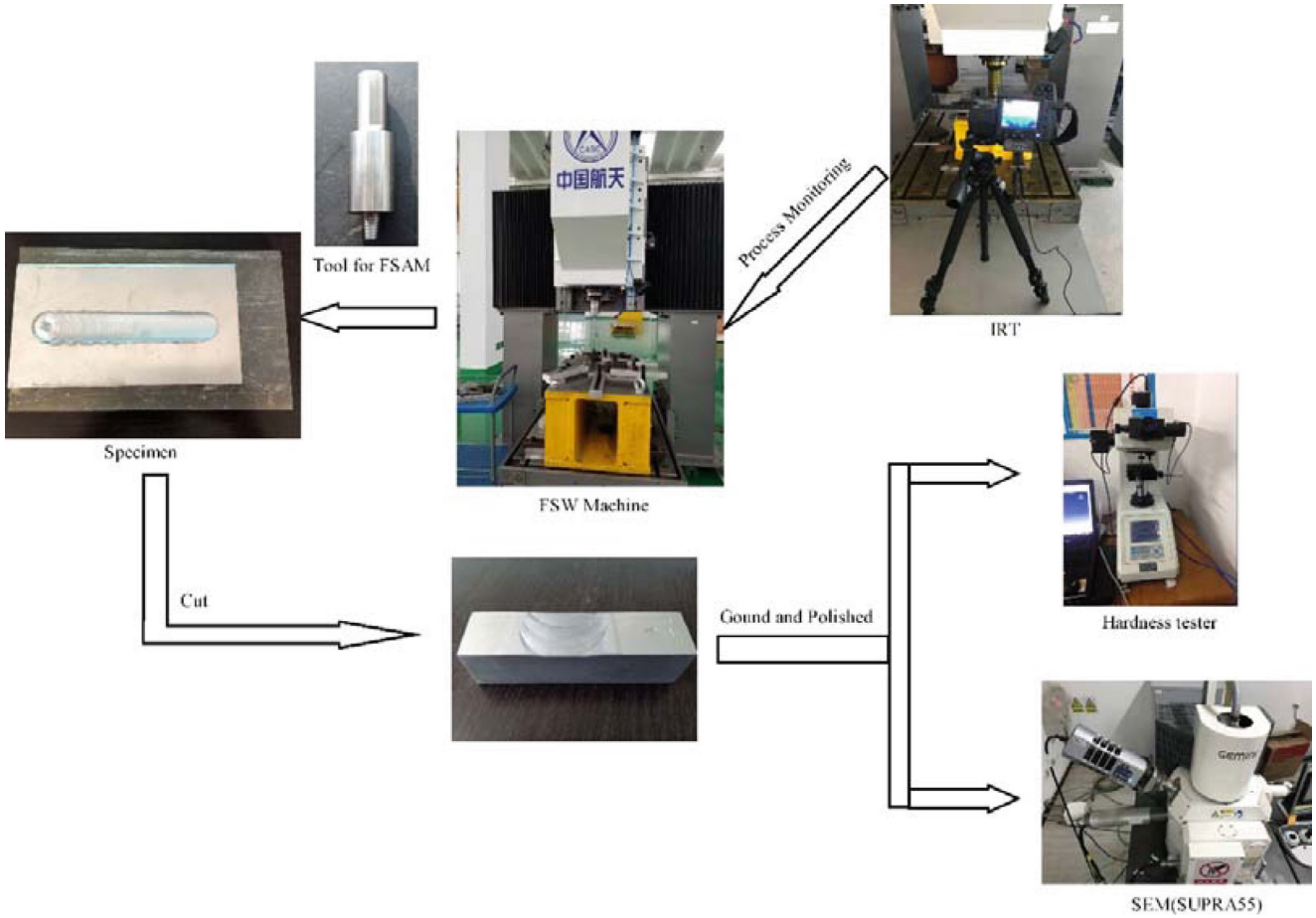
2 Experimental procedure

AA6061-T6 is selected for FSAM. The sizes of plates for FSAM are 200 × 110 × 4 mm. The substrate for FSAM is AA-6082 and the sizes are 250 × 150 × 8 mm. One layer is added on the substrate as the first specimen. For another specimen, two layers are added to the substrate for comparison. The diameter of the tool shoulder is 24 mm. Conical pin is selected for FSAM. The diameters of the conical pin range from 6 mm on the bottom to 8 mm on the top of the pin. H13 steel is selected for production of welding tool as used in FSAM. The detailed experimental procedure of FSAM is shown in Fig. 1a. Figure 1b shows the schematic diagram of FSAM process. A tool rotates and then moves along the moving direction. The solid-state material flow leads to recrystallization in stirring zone. Different to FSW, the pin length is longer than the layer thickness. The workpiece is then built layer by layer along building direction. FSW machine is used for FSAM. The rotating speed of the tool is 1000 rev/min and the transverse speed is 100 mm/min. Infrared radiation thermometer (IRT) is used for process monitoring of temperature in FSAM process. The obtained specimens are cut and then ground and polished with a diamond paste of 0.5 μm. Then, the samples were etched in the Keller's reagent for 200 s. Scanning electron microscope (SEM) is used for observation and measurement of grain sizes. Vickers hardness tester (VHT) is used for the measurement of hardness distribution on the cross section of FSAM specimens. The specimen with two layers is compared with the one with one layer to show the re-stirring and re-heating effects.

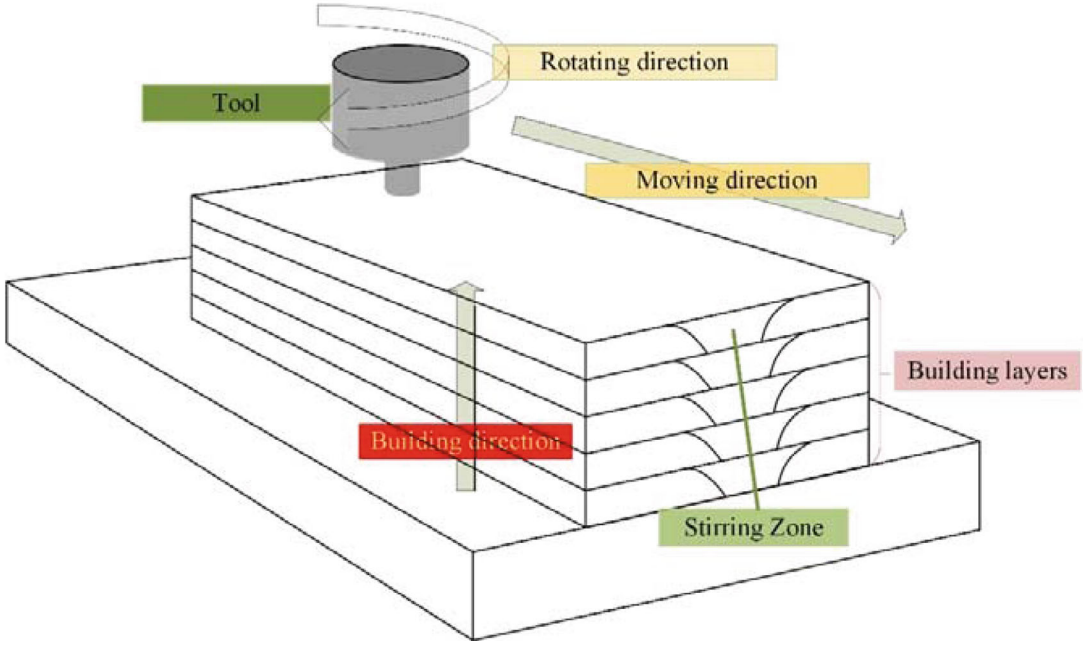
3 Numerical models

3.1 Moving heat source model

In FSAM, the generated heat comes from frictional and plastic dissipations, in which the friction takes the main contribution. Zhang et al. [19] studied the effect of plate thickness on ratio of frictional and plastic dissipations. Zhang and Zhang [20] investigated effect of process parameters on frictional and plastic contributions to temperatures. Zhang and Zhang [21] studied the transverse speed on temperature rises and the mechanism for formation of defects. It was found that the contribution from plastic dissipation is smaller than friction. So, a moving heat source calculated from friction between the welding tool and the work piece is used to simulate the temperature rises in FSAM. This model has been successfully applied to FSW. Yan et al. [22] used this model to predict temperature and then investigate the



(a) Experimental procedure of FSAM



(b) Schematic diagram of FSAM

Fig. 1 Experimental procedure and schematic diagram of FSAM

temperature-induced distortion by a sequentially coupled model. Zhang et al. [23] used this model to investigate the effect of temperature-induced residual stress on fatigue life of joint. Zhang et al. [24] used this model to further study the effect of friction stir weld size on final distortions. So, this moving heat source model is also adopted here for temperature predictions and for comparison with experimental test. The heat source is simplified to be a surface heat input,

$$P_s = \eta \frac{2}{3} \pi \mu p \omega \left(r_s^3 + 3 l_p r_p^2 \right) \tag{1}$$

where η is the heat fraction flowing into welding plate on tool-plate contact surface and is taken as 0.39 according to Nandan et al. [25]. μ is the frictional coefficient and is taken as 0.5. P is the contact pressure along axial direction. ω is the rotational speed. l_p is the length of pin. r_s and r_p are the radii of shoulder and pin and are 12 mm and 4 mm, respectively.

The sizes of the welds and the substrate in the finite element model are identically the same with the ones used in experiment. The process parameters are also selected to be the same with experimental settings. The minimum element size is 2 mm. A total of 7200 8-node linear brick elements and 14,570 nodes are used for meshing of the substrate. A total of 11,000 elements and 16,968 nodes are used for meshing of each layer. The finite element model is shown in Fig. 2. Birth-death element strategy is used for simulation of the added *layers*. The thermal parameters of the selected material are considered to be dependent on temperatures, as shown in reference [24]. Convective boundary condition is selected,

$$k \frac{\partial T}{\partial n} = h(T - T_a) \tag{2}$$

where k is the thermal conductivity, h is the convection coefficient, and T_a is the ambient temperature.

ABAQUS is selected for simulation of the moving heat source. The obtained temperature histories are used for further simulations on grains and precipitates.

3.2 Monte Carlo model for recrystallization

The flow chart for Monte Carlo simulation of grain evolutions is shown in Fig. 3. Different to previous work on Monte Carlo simulation in friction stir welding [11], the precipitate variations in the FSAM process, which are calculated from a developed precipitate evolution model, are considered in current work. Real square area in SZ is represented by $N \times N$ lattice matrix. Each lattice point is given a number $1 \sim q_{\max}$ to describe its crystallographic orientations. A special orientation $q_{\max} + 1$ is assigned to lattice sites *which* represent the precipitations. The energy of the system in the Monte Carlo simulation is described as

$$E = -J \sum_{i=1}^{m} \left(\delta_{q_i q_j} - 1 \right) \tag{3}$$

where J is the lattice energy density. $\delta_{q_i q_j}$ is the Kronecker delta function. m is the total number of sites near one lattice point.

During each reorientation step, a lattice site from nucleation point can be chosen for grain coarsening. The nucleation rate is as follows [26],

$$\dot{n} = N_0 \dot{\bar{\varepsilon}} \exp\left(-\frac{Q}{RT} \right) \tag{4}$$

where N_0 is the constant and is taken as 10^{24}/s/m^3. Q is the activity energy. R is the gas constant. T is the temperature. The strain rate can be directly affected by the rotating speed [27] and the welding tool [28]. It is calculated according to the size of pin and the rotating speed [29],

$$\dot{\bar{\varepsilon}} = \frac{\omega \pi r_e}{L_e} \tag{5}$$

where r_e and L_e are the effective radius and depth of the dynamically recrystallized zone (SZ). r_e is represented by 0.78 of the radius of stirring zone and L_e is assumed to be the length of the pin.

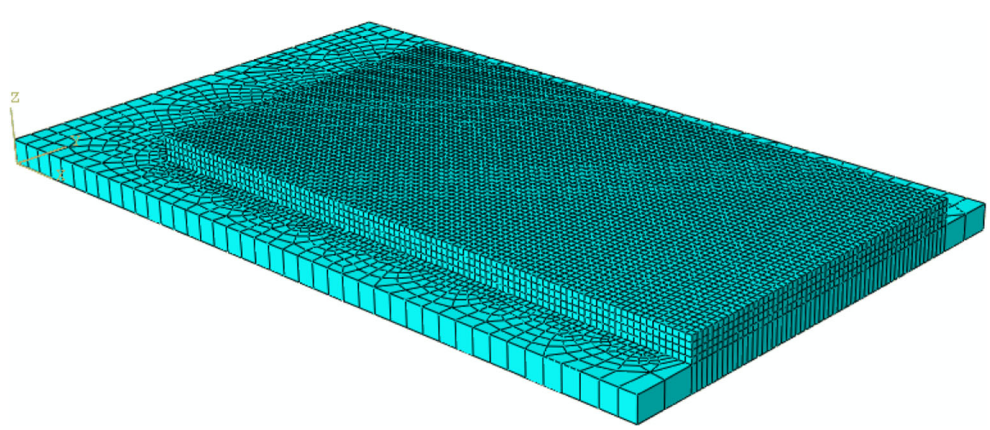


Fig. 2 Finite element model

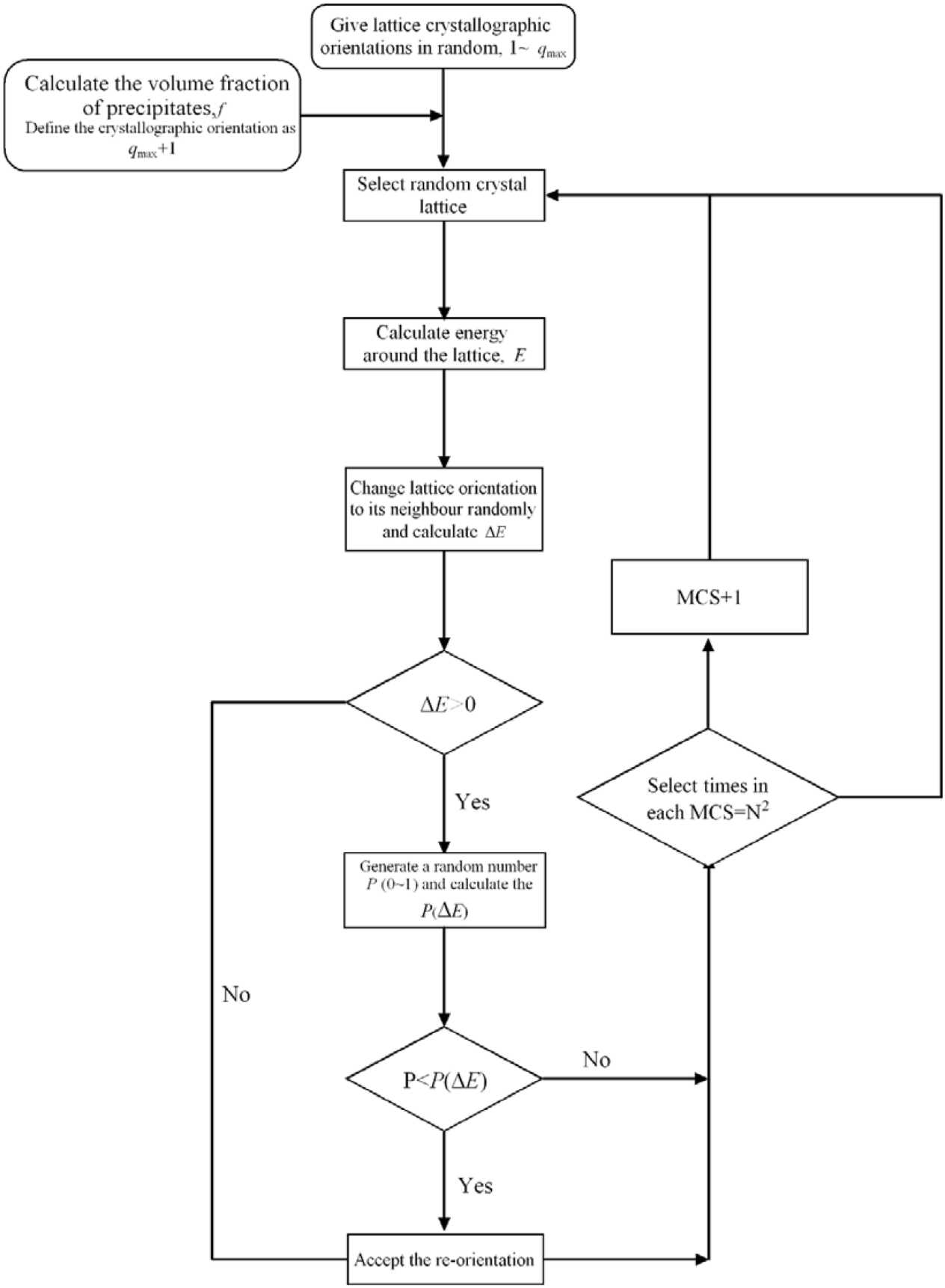


Fig. 3 Flow chart for Monte Carlo model

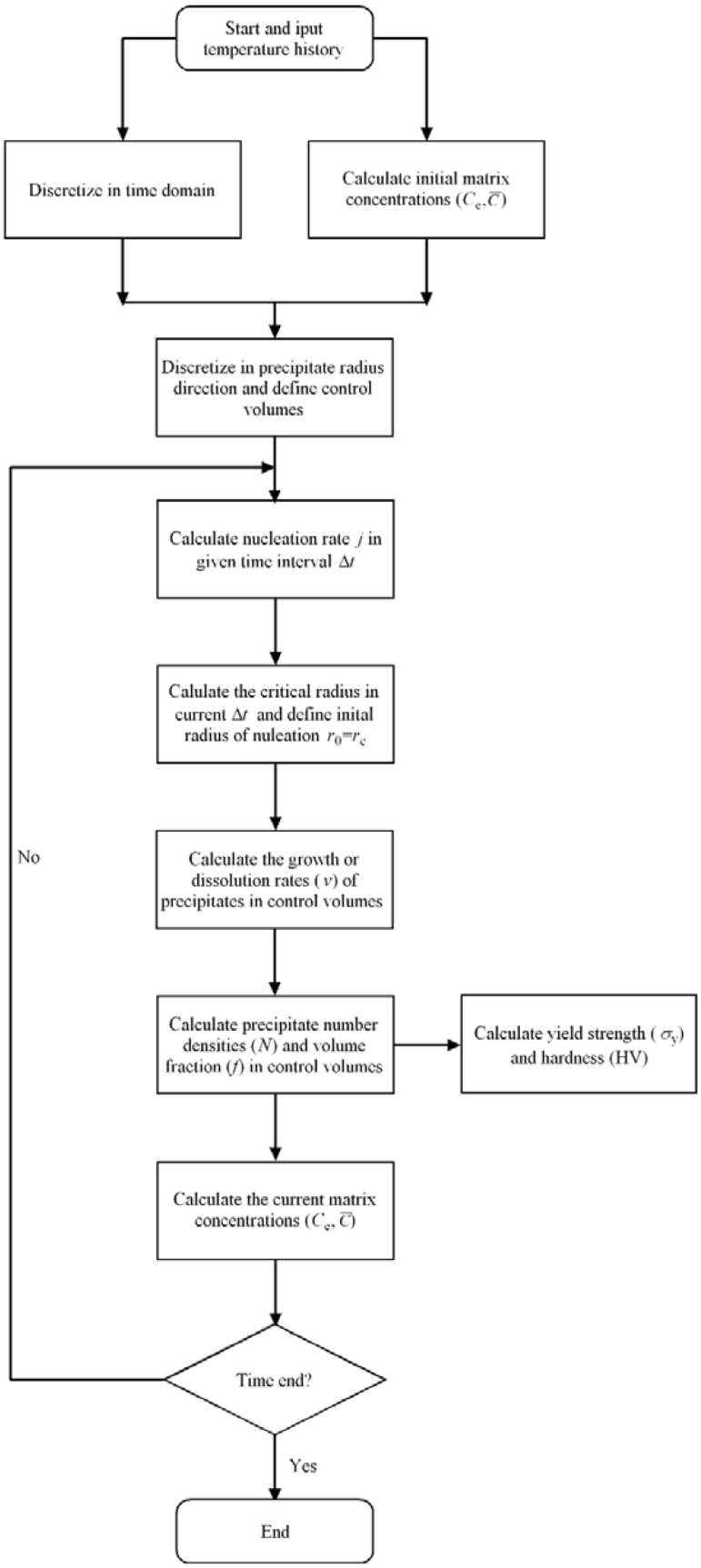


Fig. 4 Flow chart for precipitate evolution model

The nucleation sites are randomly distributed into the matrix. The orientation of the randomly selected site is changed to be equal to its neighbors. If the energy change is smaller than 0, the re-orientation is accepted. If the energy change is not smaller than 0, the re-orientation is accepted according to the following probability,

$$p = \begin{cases} 1, & \Delta E \le 0 \\ e^{\frac{\Delta E}{k_B T}}, & \Delta E > 0 \end{cases} \tag{6}$$

where ΔE is the change of grain-boundary energy due to reorientation, k_B is the Boltzmann constant, and T is the temperature. A relation between MCS and real temperature is established,

$$(MCS)^{(n+1)n_1} = \left(\frac{L_0}{K_1 l}\right)^{n+1} + \frac{(n+1)\alpha C_1^n}{(K_1 l)^{n+1}} \sum \left[\exp^n\left(-\frac{Q}{RT_i}\right) t_i\right] \tag{7}$$

where $C1$ can be described by the following equation,

$$C_1 = \frac{2A\gamma Z V_m^2}{N_a^2 h} \exp\left(\frac{\Delta S_f}{R}\right) \tag{8}$$

where A, Z, V_m, ΔS_f, R, N_a, γ, Q, α, n, and h are summarized in reference [11].

3.3 Precipitate evolutions and predictions of mechanical properties

The flow chart for calculations of precipitates and mechanical properties is shown in Fig. 4.

The nucleation rate for precipitate in FSAM can be written as [30],

$$j = j_0 \exp\left[-\left(\frac{A_0}{RT}\right)^3 \left(\frac{1}{\ln\left(\overline{C}/C_e\right)}\right)^2\right] \exp\left(-\frac{Q_d}{RT}\right) \tag{9}$$

where j_0(#/(m^3·s)) is the material coefficient, which is the pre-exponential term to the nucleation rate j and taken as 3.07×10^{36}. R (8.314 J K^{-1} mol^{-1}) is the universal molar gas constant. Q_d (J mol^{-1}) is the diffusion activation energy of element Mg in aluminum matrix, assuming that Mg is controlling the growth or dissolution of precipitation. $\overline{C}$ (wt.%) is the mean concentration of Mg in matrix. A_0 (J mol^{-1}) is the parameter related to the energy barrier for nucleation and its value is taken as 18 kJ/mol. C_e (wt.%) is the equilibrium solute concentration at the precipitate/matrix interface,

$$C_e = C_s \exp\left(-\frac{Q_s}{RT}\right) \tag{10}$$

where C_s is the pre-exponential term in simplified expression for Mg equilibrium concentration, which is taken as 290 wt.%. Q_s is the solvus boundary enthalpy, which is taken as 41 kJ/mol.

The growth rate of the precipitated particle is determined by reference [31],

$$v = \frac{dr}{dt} = \frac{\overline{C} - C_i}{C_p - C_i} \frac{D}{r} \tag{11}$$

where C_i (wt.%) represents the solute at particle/matrix interface. C_p is the concentration of element Mg inside the precipitates. It is assumed that Mg_5Si_3 is the only stoichiometric composition in the precipitates. D (m^2 s^{-1}) is the diffusion coefficient of solute in matrix. r is the particle radius.

According to the Gibbs-Thomson equation, the solute at particle/matrix interface can be expressed as,

$$C_i = C_e \exp\left(\frac{2\gamma V_m}{rRT}\right) \tag{12}$$

where γ (J m^{-2}) is the particle/matrix interfacial energy, 0.26 J/m^2, and V_m (m^3 mol^{-1}) is the molar volume of precipitation, 7.62×10^{-5} m^3/mol.

The growth rate of the precipitates is zero when $\overline{C} = C_e$. So, the critical radius of the precipitated particles can be obtained,

$$r_c = \frac{2\gamma V_m}{RT} \left(\ln\left(\frac{\overline{C}}{C_e}\right)\right)^{-1} \tag{13}$$

Then, a number density of $j\Delta t$ nucleated particles is added during each time step, where the radius of a nucleated particle equals 1.05 r_c. Because a new nucleated particle only can grow if the radius is a little bit larger than the critical radius. The Mg solutes are either in precipitates or matrix of alloy. Its mass balance can be expressed as following equation,

$$\overline{C} = C_0 - \left(C_p - \overline{C}\right) f \tag{14}$$

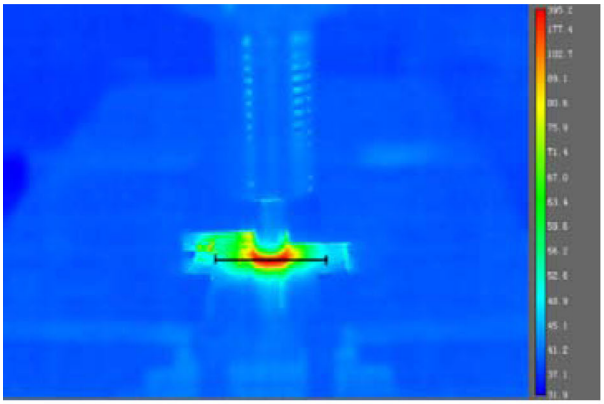
where C_0 is the initial concentration of Mg in aluminum alloy.

The model for precipitates is discretized in radius direction. The volume fraction of precipitates is then calculated as,

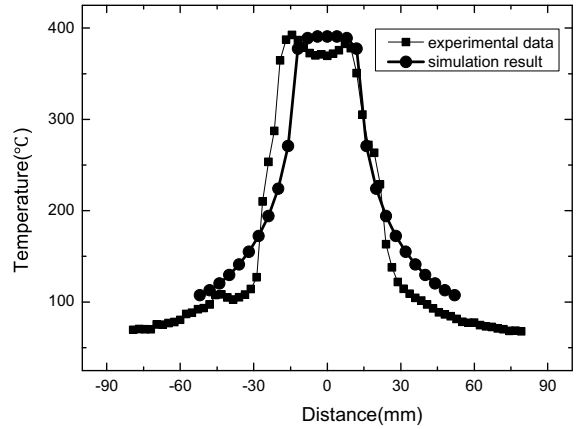
$$f = \frac{4}{3}\pi r_i^3 N_i = \left(C_0 - \overline{C}\right) / \left(C_p - \overline{C}\right) \tag{15}$$

where N_i denotes the particle number of a discretized class (indexed i). The metastable and stable precipitates (β'', β', and β) are assumed to be spherical.

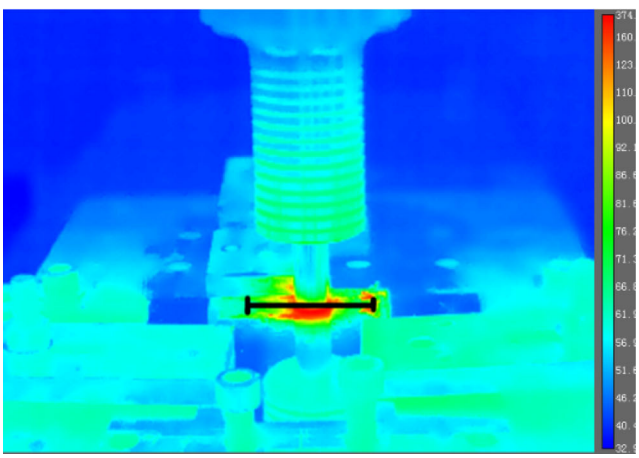
The calculated volume fraction is transferred to Monte Carlo model for simulation of recrystallization.



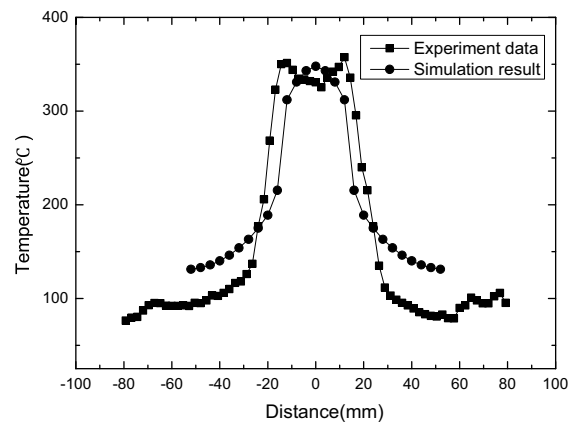
(a) Temperature in 1st layer from IRT system



(b) Comparison of temperature in 1st layer



(c) Temperature in 2nd layer from IRT system



(d) Comparison of temperature in 2nd layer

Fig. 5 Comparison of temperature fields between experiment and numerical model

The yield strength of the friction stir weld can be divided into three parts,

$$\sigma_y = \sigma_0 + \sigma_{ss} + \sigma_p \tag{16}$$

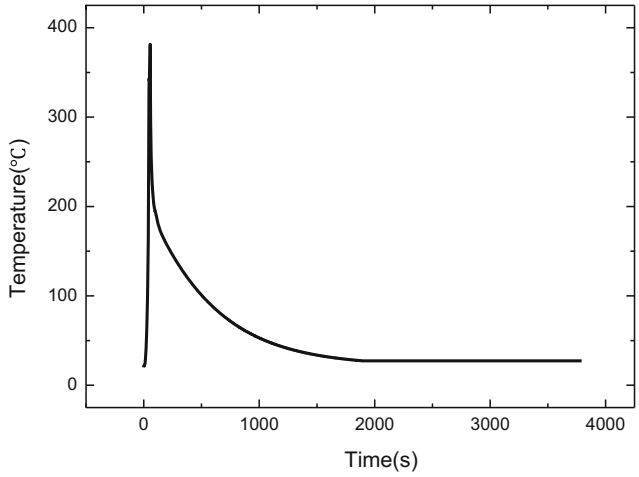
where σ_{ss} is the additional contribution from solid solution and σ_p from precipitates. σ_0 is the intrinsic yield strength of pure aluminum including the contribution from the dislocation density.

The solute hardening due to the interstitials is computed by,

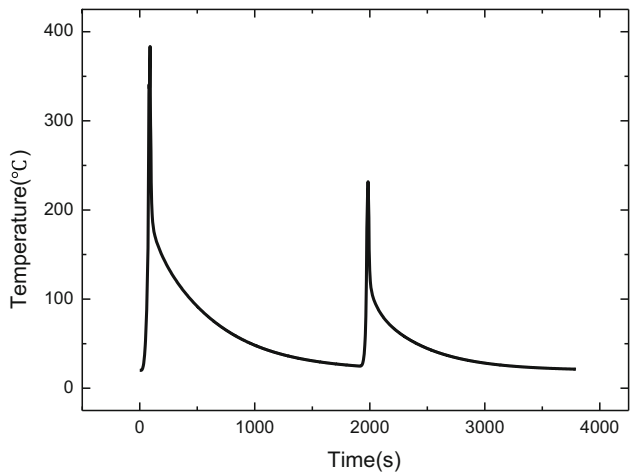
$$\sigma_{ss} = \sum_{i=1}^{3} k_i C_i^{2/3} \tag{17}$$

k_i is the scaling factor for alloying element i and C_i is its concentration. The solute elements for solid solution are Mg, Si, and Cu, indexed $i = 1$, 2, and 3. $k_1 = 66.3$ (MPa/wt%$^{2/3}$), $k_2 = 29$ (MPa/wt%$^{2/3}$), and $k_3 = 46.4$ (MPa/wt%$^{2/3}$). The precipitate hardening is given by,

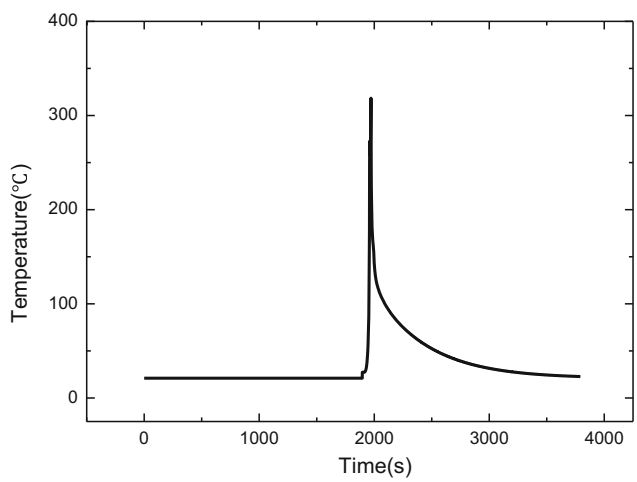
$$\sigma_p = \frac{M}{b\bar{r}} \left(2\beta G b^2\right)^{-1/2} \left(\frac{3f}{2\pi}\right)^{1/2} \left(\frac{\sum_i N_i F_i}{\sum_i N_i}\right)^{3/2} \tag{18}$$



(a) 1-layer specimen



(b) first layer for 2-layers specimen



(c) second layer for 2-layers specimen

Fig. 6 Temperature histories for the two specimens

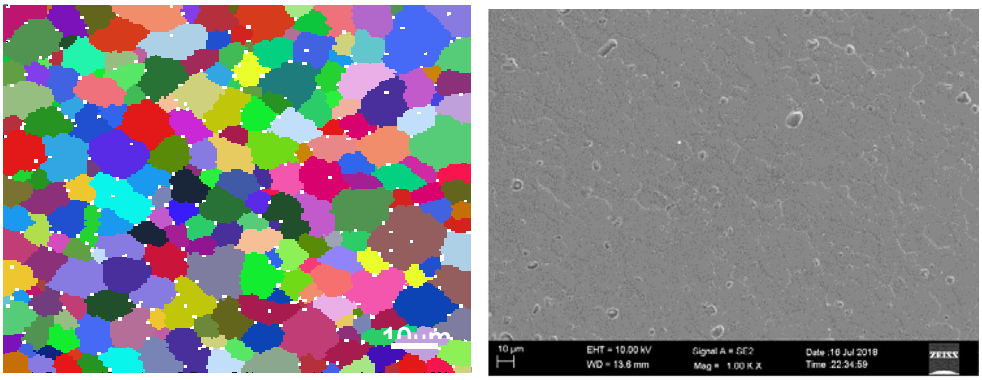
where $\beta$is the constant, F_i is the obstacle strength of the corresponding size in given discretized class and is related to the particle size. The calculation of F_i can be referred to reference [14].

4 Results and discussions

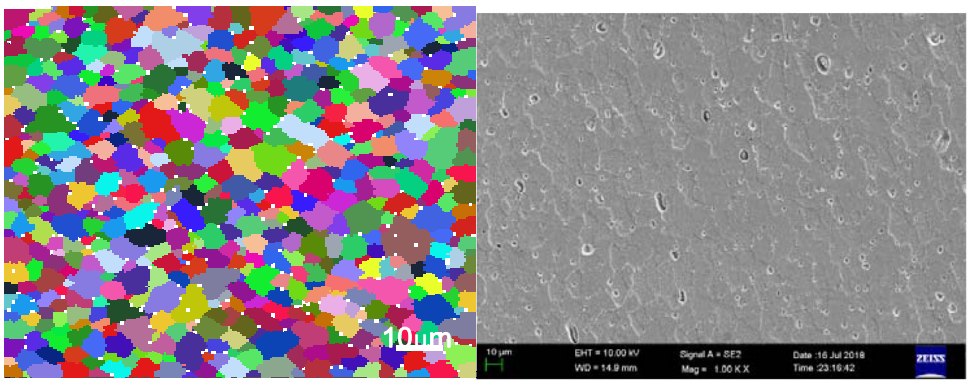
The temperature field in FSAM is measured by Infrared Radiation Image (IRT) system. The observed temperatures of 1st layer and 2nd layer from experiment in 1000 rev/min and 100 mm/min are shown in Fig. 5a, c. The extracted data for 1st layer and 2nd layer along the direction perpendicular to the welding line *are* compared to the numerical model, as shown in Fig. 5b, d. The maximum temperature of the 1st layer from experiment is 392.4 °C and is 390.7 °C in numerical model on the selected path. The error for peak temperature is 0.4%. The maximum temperature of the 2nd layer from experiment is 351.1 °C and is 347.7 °C in numerical model on the selected path. The error for peak temperature is 0.9%. The numerical model can be fitted well with the experimental measured data. This serves as the basis for further simulations on microstructures and mechanical properties.

Two specimens, one-layer specimen and two-layer specimen, are chosen for further numerical simulation and comparison. The temperature histories at selected points on the centerline and being 2 mm from the top surface are exacted for comparison, as shown in Fig. 6. The temperature curve on the first layer in two-layer specimen is initially the same with the one-layer specimen. But due to the addition of next layer in this two-layer specimen, the first layer in two-layer specimen is re-heated. The re-heating for layers is a special feature for additive manufacturing technology. The temperature data on selected point is exacted for further investigations on how the re-heating affects the final microstructures and mechanical properties. Due to the fact that the heat in FSAM is mainly generated on the shoulder-plate interface, the temperature in re-heating process becomes smaller than the heating process caused by the increased size of specimen. The maximum value is decreased from 385 °C in heating process to 231.7 °C in re-heating process. The comparison of temperature histories in different layers shows that the maximum temperature is decreased from 385 °C in first layer to 324.1 °C in the second layer. The peak temperature of new added layer is decreased with the increase of build height in FSAM. Experimental observation from IRI system also shows this point.

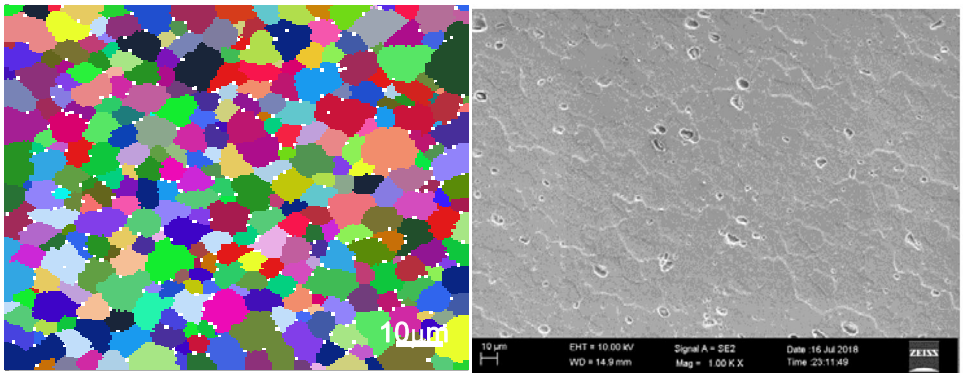
The grain sizes and morphologies are predicted by Monte Carlo model with consideration of precipitate volume fractions calculated from precipitate evolution model. Similar to multi-layered FSW [32] and self-riveting friction stir lap joints [33], the grains in stirring zone can be refined. But the existing re-stirring and re-heating in FSAM lead to new features. The predicted microstructures are compared to experimental ones



(a) 1-layer specimen



(b) first layer for 2-layers specimen



(c) second layer for 2-layers specimen

Fig. 7 Comparison of grains between experiment and numerical model

observed by SEM, as shown in Fig. 7. In comparison with the one-layer specimen, the average grain size in the first layer in 2-layers specimen is decreased to 9.65 μm from 15.5 μm in numerical model and decreased to 10 μm from 15.9 μm in experiment. The first layer in two-layer specimen is re-heated, as indicted by temperature curves. But at the same time, the material in this layer is re-stirred. The re-stirring leads to a new recrystallization. The recrystallization with lower temperature is the real reason for the decrease of grain sizes in the first layer in two-layer specimen. Although the maximum temperature in the second layer is slightly lower than the one in the first layer in two-layer specimen, it is higher than the one in the re-heating process. So, the grain size in the second layer is still larger than the first layer. The average grain size in the first layer in two-layer specimen is 9.65 μm and 13.3 μm in the second layer in numerical model. The experimental observation is 10 μm in the first layer and 13.13 μm in the second layer. The errors are 3.5% and 1.3%, respectively. The re-stirring with lower temperature in re-heating leads to finer grain distributions in FSAM.

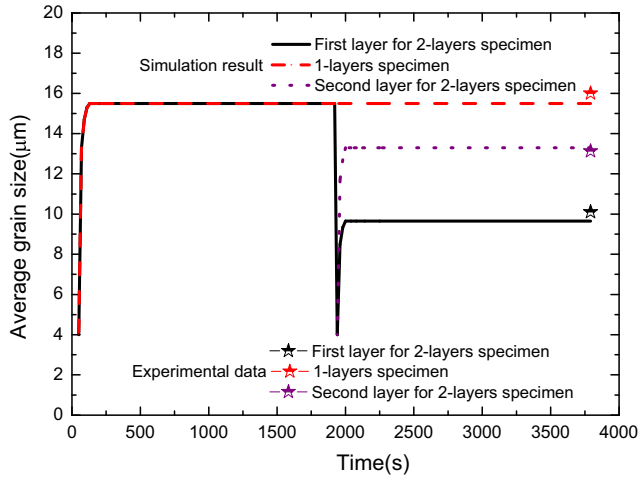


Fig. 8 Different average grain sizes in SZ on different welding layers

The grain growth histories are shown in Fig. 8. The experiment can provide the final value for the average grain size but the numerical model can provide the histories. We can easily see the recrystallization in stirring process and the new generated recrystallization in the re-stirring process.

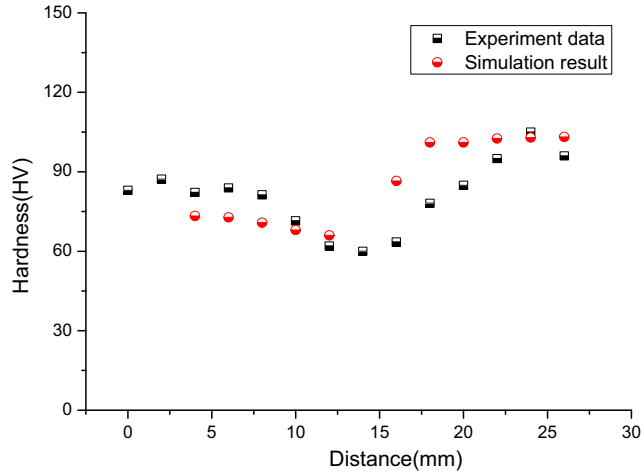
The relation between the hardness and the yield strength can be given as,

$$HV = 0.33\sigma_y + 16 \tag{19}$$

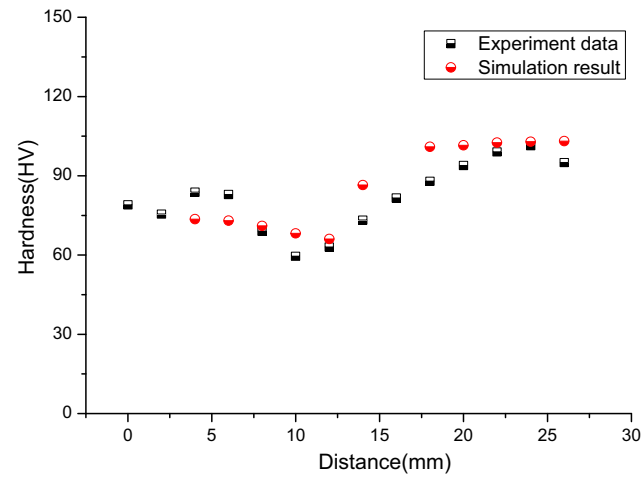
The calculated hardness can be directly compared to experimental test data by Vickers hardness tester, as shown in Fig. 9. The points are selected near the middle surface. The hardness measured by VHT in the stirring zone and thermomechnical affected zone are similar to the one-layer specimen. The predicted values can be fitted well with the experimental ones. But for Fig. 9c, there are differences between experimental and numerical data. In numerical model, the position of tool is precise when the new layer is added. However, there is small shift for the tool position when new layer is added in experiment. This is the reason for the differences between numerical model and experimental data. But the comparison can still show the validity of the proposed integrated numerical model.

To show the variations of mechanical properties, we take point being 6 mm away from centerline for example in the numerical model of two-layer specimen, as shown in Fig. 10 and Fig. 11. Yield strength can be divided into three parts according to in Eq. (16). Precipitate hardening takes the main contribution to the yield strength. For the first layer, σ_p takes 59% contribution to yield strength. For the second layer, σ_p takes 63% contribution to yield strength.

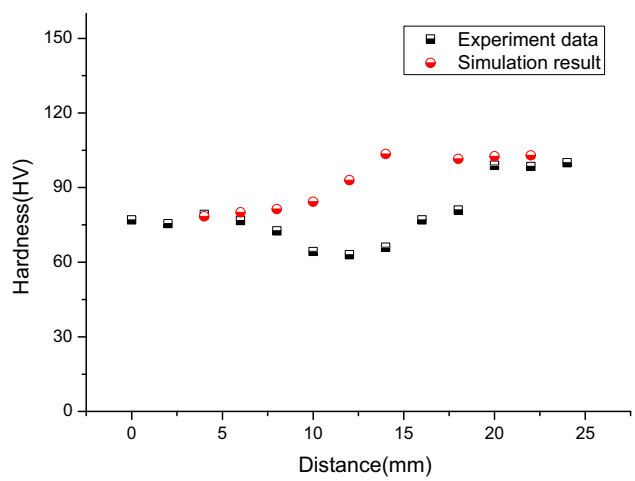
The volume fraction and mean radii of precipitates are shown in Fig. 12 for further explanations. In the first layer of the two-layer specimen, the particle number of precipitates (Mg_2Si) is decreased from $2.40231 \times 10^{22}/m^3$



(a) 1-layer specimen



(b) first layer for 2-layers specimen



(c) second layer for 2-layers specimen

Fig. 9 Comparison of hardness between experiment and numerical model

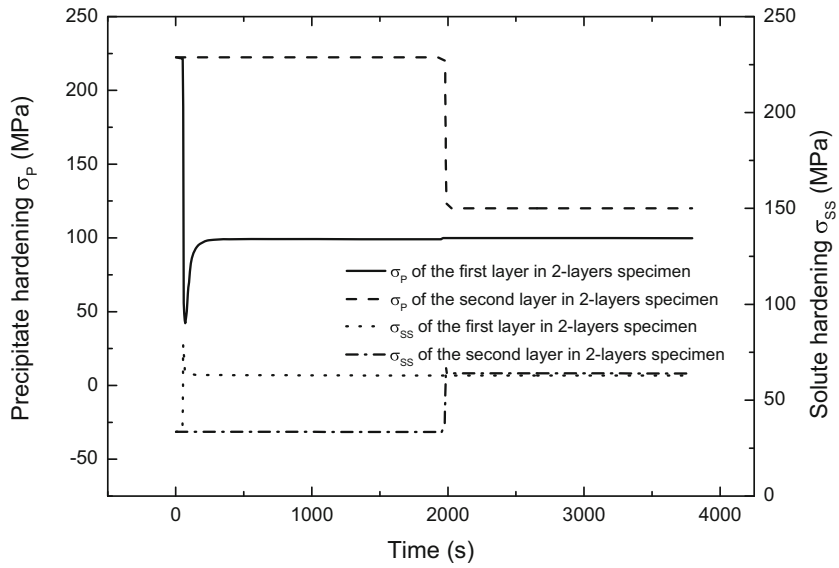


Fig. 10 Variations of precipitate hardening and solute hardening in two-layer specimen

in base metal to $2.12863 \times 10^{20}/m^3$ at the peak temperature. Then, Mg_2Si is precipitated in the following cooling process and the particle number starts to be increased to $2.10378 \times 10^{21}/m^3$. Meanwhile, the mean radii of precipitates are changed in the heating process. In the heating process, the mean radius of precipitates is increased from 4.69484 to 19.5224 nm. In the cooling process, the mean radius of precipitates is decreased to 4.09552 nm with the increase of particle number of precipitates. The increase of precipitate with smaller mean radius can be beneficial to the enhancement of the material. In the re-heating process, the particle number of precipitates is increased to $2.21116 \times 10^{21}/m^3$ for the first layer in the re-heating process. This value is very similar to the status before re-heating. The mean radius is slightly decreased to 4.0166 nm. The decrease of mean radii and the increase of particle number density is the real reason for the increase of mechanical properties after re-stirring and re-heating. For the second layer, there is no re-heating process. In the heating process, the particle number of precipitates is decreased from $2.40231 \times 10^{22}/m^3$ in base metal to $3.53752 \times 10^{21}/m^3$, which is higher than the first layer. The mean radius of precipitates increased from 4.69484 to 6.95327 nm. In the following cooling process after peak temperature, the particle number of precipitates is $3.67292 \times 10^{21}/m^3$, which is higher than first layer. The mean radius is decreased to 6.83551 nm. In current case, the mean radius and particle number of precipitates are both increased along the building direction. So, the hardness is increased from 73.02 HV in the first layer to 80.06 HV in the second layer (as shown in Fig. 9) as well as the yield strength from 172.79 to 194.12 MPa as calculated by Eq. (19). The increase of precipitate radius and particle number is the direct reason for the increase of mechanical property along the building direction. To reveal this mechanism, the distributions of particle sizes of both two layers are further revealed in Fig. 13. In comparison with the first layer, the precipitates are coarsening in the second layer, which is the reason for increase of hardness and yield strength along the building direction.

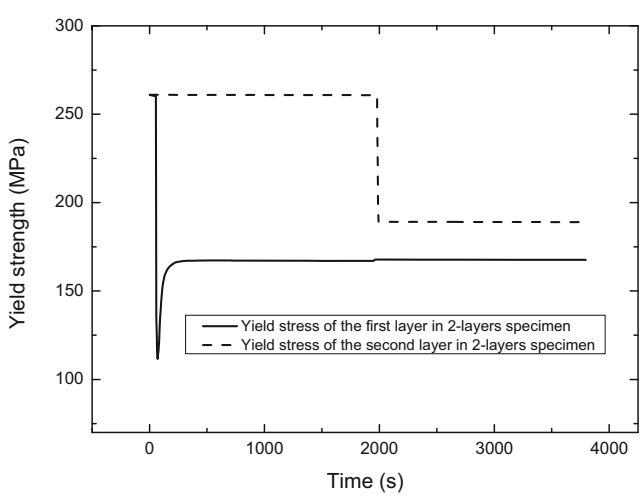


Fig. 11 Variations of yield strength in two-layer specimen

To study the effect of re-heating on precipitates, the variations of precipitate mean radii of one-layer specimen and 1st layer of two-layer specimen are compared in Fig. 14. We further take point being 4 mm away from centerline for comparison. There is no re-heating stage for the one-layer specimen. So, the mean radius keeps constant in this period. But for 1st layer of two-layer specimen, the material is re-heated and the mean radius is decreased to 3.86464 nm from 3.94244 nm. With similar volume fraction, the decrease of mean radius can lead to increase of mechanical properties. The hardness is

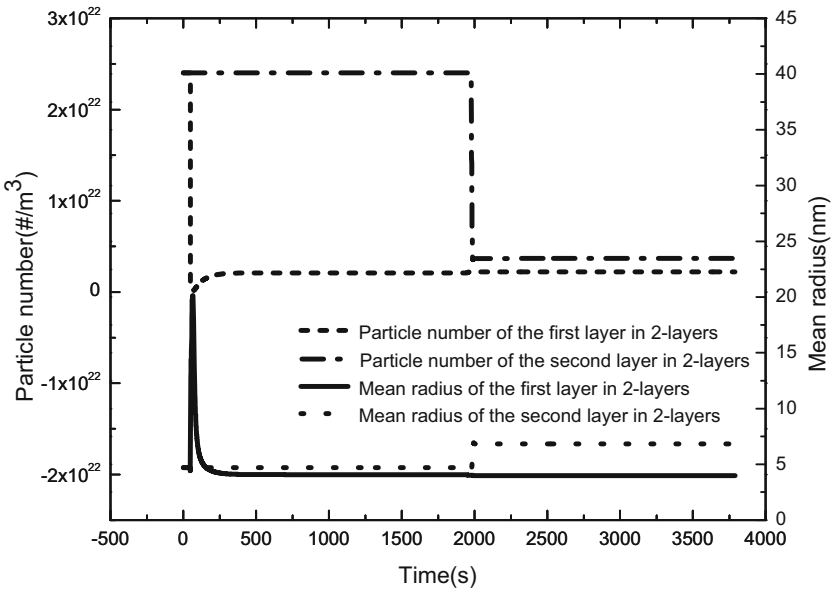


Fig. 12 Variations of volume fraction and mean radii of precipitates in two-layer specimen in 1000 rev/min and 100 mm/s

increased from 73.34721 to 73.57906 HV (as shown in Fig. 9) as well as the increase of yield strength from 173.78 to 174.482 MPa, as calculated by Eq. (19). The hardness, as well as yield strength, is increased by the re-heating and re-stirring effects in FSAM.

The effects of the transverse and rotational speeds on microstructure and hardness are detailed studied. The increase of the transverse speeds and decrease of rotation speeds can lead to the decrease of peak temperature in the stirring zone as shown in Fig. 15. The selected points are 2 mm below the upper surface of the weld center. When the transverse speed is increased from 60 to 100 mm/min, the peak temperature of 1st layer is decreased from 433 to 385 °C. When the rotation speed is decreased from 1200 to 1000 mm/min, the peak temperature of 1st layer is decreased from 418 to 385 °C. The temperature distributions are shown in Fig. 16. The peak temperatures of selected points are slightly lower than upper surface.

The grain growth histories are shown in Fig. 17. After the first layer welded, the grain size on 1st layer in 1000 rpm to 100 mm/s case is 15.5 μm, 17 μm in 1200 rpm to 100 mm/s case, and 18 μm in 1000 rpm to 60 mm/s case. When the 2nd layer is added, the material in 1st layer is re-stirred. The grain size on 1st layer in 1000 rpm to 100 mm/s case is reduced to 9.65 μm, 9.8 μm in 1200 rpm to 100 mm/s case, and 10 μm in 1000 rpm to 60 mm/s case. When the third layer is added, the grain size in the first layer of the stirring zone does not further change. As shown in Fig. 17, the recrystallization with lower

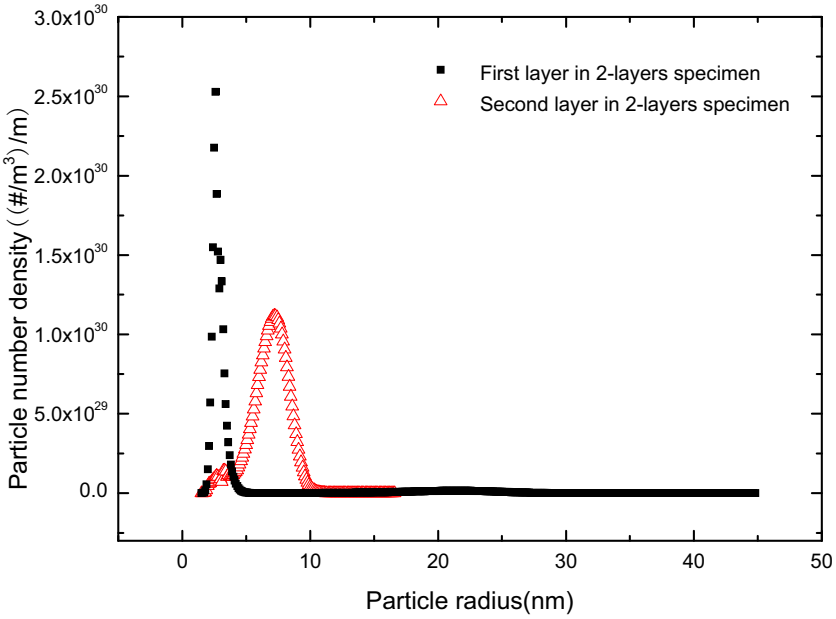


Fig. 13 Particle size distribution on radius

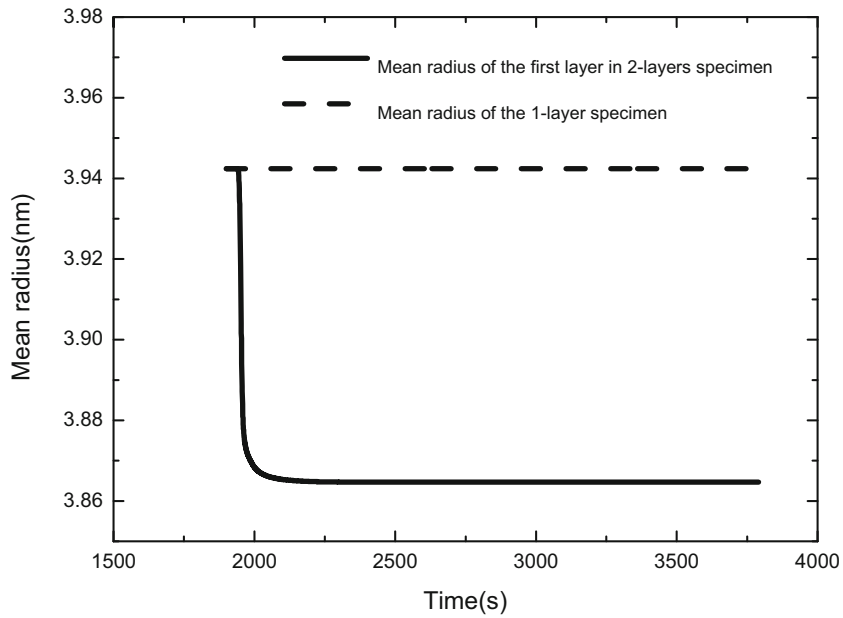


Fig. 14 Variations of mean radii of one-layer specimen and 1st layer of two-layer specimen

temperature can decrease the grain size and improve quality of joint and reduce the difference of microstructure in different cases.

With the increase of added layers, the re-heating time on the first layer is increased. Hardness and yield stress on the first layer can be also affected by such re-heating. The increase of the transverse speed or decrease of rotational speed can lead to the decrease of hardness and the yield strength in the stirring zone. The increase of transverse speed or decrease of rotational speed usually leads to decrease of peak temperature in the FSAM process, as shown in Fig. 16. This phenomenon, caused by interface behavior [34], is also observed in a conventional friction stir welding [35, 36]. But different to friction stir welding, new layers are continuously added. The mechanical properties can be affected by the new added layers. So, we take point being 5 mm away from the centerline in the 1st layer for comparison of the re-heating effects caused by the new added layers as shown in Fig. 18. When the transverse speed is 60 mm/min and the rotational speeds is 1000 rev/min, the hardness of the 1st layer is 64.8 HV after the 1st layer is added, 68.6 HV after the 2nd layer is added, 69.5 HV after the 3rd layer is added, 69.7 HV after the 4th layer is added, and 69.7 HV after the 5th layer is added. When the transverse speed is increased to 100 mm/min, the hardness of the

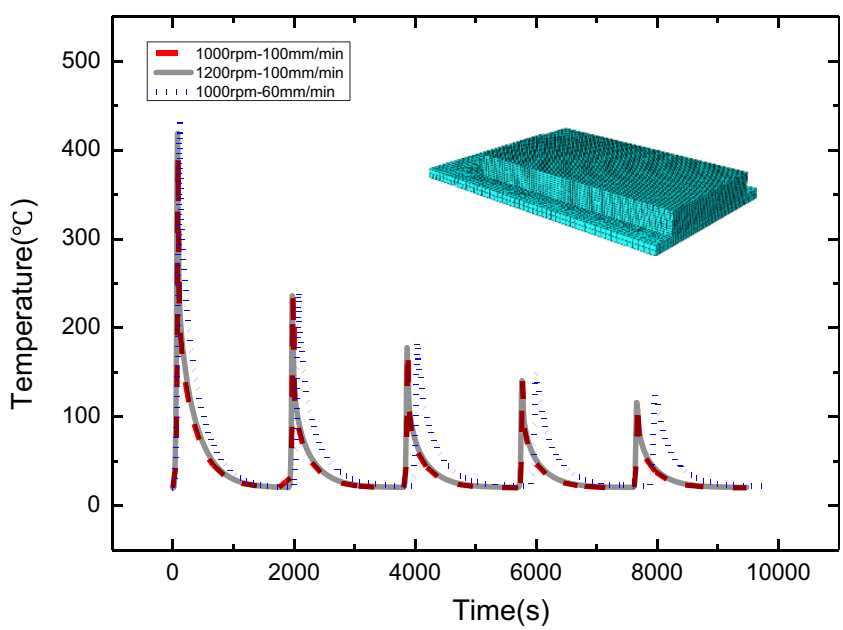
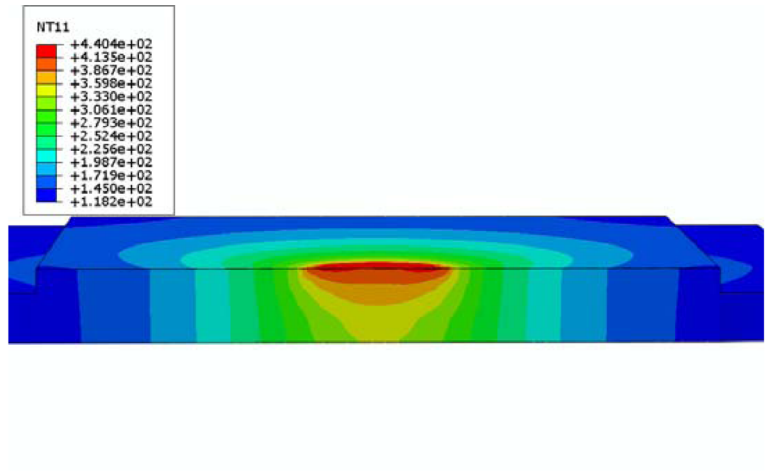
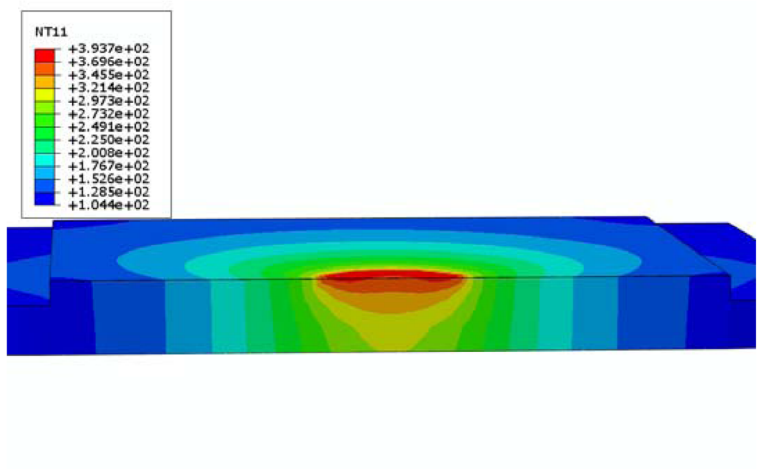


Fig. 15 Temperature histories for the five specimens

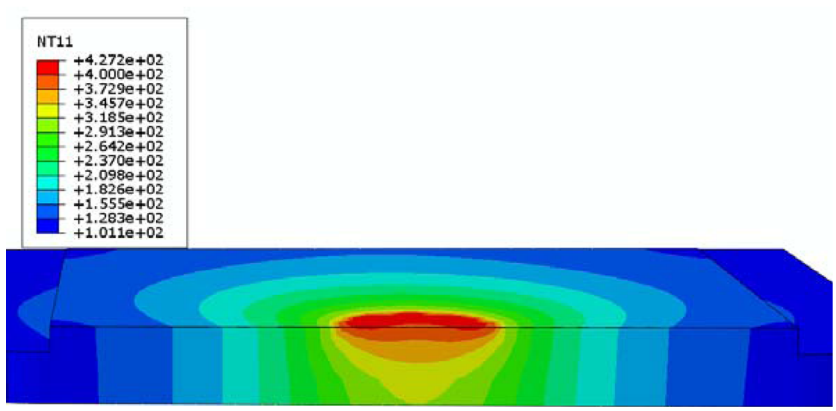
Fig. 16 Temperature distribution of 1st layer for the five specimens



(a) 1000rpm, 60mm/min



(b) 1000rpm, 100mm/min



(c) 1200rpm, 100mm/min

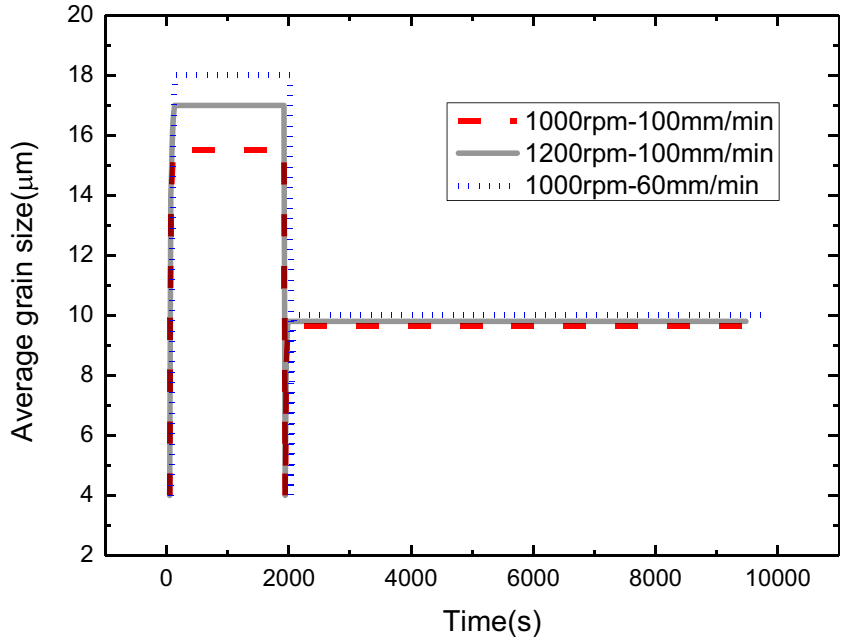


Fig. 17 Average grain sizes for the five-layer specimens

1st layer is 61.0 HV after the 1st layer is added, 61.5 HV after the 2nd layer is added, 61.7 HV after the 3rd layer is added, 61.7 HV after the 4th layer is added, and 61.7 HV after the 5th layer is added. When the rotational speed is further changed to 1200 rev/min, the hardness of the 1st layer is 63.5 HV after the 1st layer is added, 65.4 HV after the 2nd layer is added, 65.7 HV after the 3rd layer is added, 65.8 HV after the 4th layer is added, and 65.8 HV after the 5th layer is added. After 2–3 layers, the mechanical properties of formed layers are not further affected by new added layers.

5 Conclusions

(1) An integrated model for prediction of microstructures and mechanical properties in FSAM is proposed with experimental validation.
(2) The peak temperature of new added layer is decreased with the increase of build height in FSAM.
(3) The re-stirring with lower temperature in re-heating leads to finer grain distributions in FSAM.
(4) The hardness, as well as yield strength, is increased by the re-heating and re-stirring effects in FSAM.

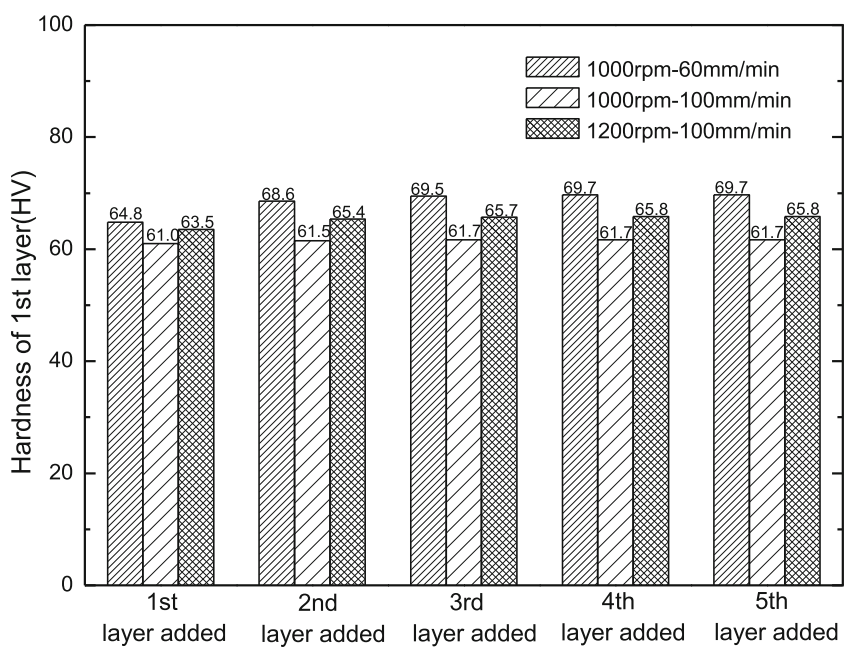


Fig. 18 Effect of transverse speeds and rotational speeds on hardness

(5) The hardness, as well as yield strength, is increased along the building direction in FSAM caused by the lower peak temperature.
(6) After 2–3 layers, the mechanical properties of the formed layers are not further affected by new added layers.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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